

oscillations will be affected as there will be much more damping than before.

- 8 (a)** Resistance R is a property of a sample (or electrical component) where it is defined as the ratio of the potential difference V applied across it to the current I that flows through it. $R = \frac{V}{I}$.

Resistivity ρ is a property of the material and is a measure of the resistance of the material to the flow of an electric current. $R = \rho \frac{L}{A}$ where L is the length of the component, A is the cross-sectional area of the component and R is the resistance of the component through which the current flows.

- (b) (i)** Using $R = \rho \frac{L}{A}$, we have

$$R = 1.7 \times 10^{-8} \times \frac{1.0}{\pi \times (0.13 \times 10^{-3})^2}$$

$$R \approx 0.32 \, \Omega$$

- (ii)** For one strand of wire of length 5.0 m, $R = 0.32 \times 5 = 1.6 \, \Omega$.

Since the strands of wire are connected parallel to one another, we have

$$R_{cable} \approx \left(\frac{1}{1.6} \times 24 \right)^{-1}$$

$$R_{cable} \approx 0.067 \, \Omega$$

- (c) (i) 1.** Using $\mathcal{E} = I(R + r)$, and $R = 0$ for maximum current, we have

$$12.0 = I \times 3.0$$

$$I = 4.0 \, \text{A}$$

- 2.** Using $P = I^2 R$, we have

$$P = 4.0^2 \times 3.0$$

$$P = 48 \, \text{W}$$

(ii) This is to ensure that even if there is no external load ($R = 0$), the circuit will not be short-circuited.

(d) (i) $R = 1.0 \, \Omega$ and $R = 9.0 \, \Omega$

(ii) For $R = 1.0 \, \Omega$,

$$P_{in} = \frac{12^2}{1.0 + 3.0} = 36 \, \text{W}$$

$$\text{Therefore, } \eta = \frac{9}{36} \times 100\% = 25\% .$$

For $R = 9.0 \, \Omega$,

$$P_{in} = \frac{12^2}{9.0 + 3.0} = 12 \, \text{W}$$

$$\text{Therefore, } \eta = \frac{9}{12} \times 100\% = 75\%$$

(e) From (d), it is observed that when r is larger than R , the efficiency is lesser. Hence, r should be chosen such that it is smaller than the component's fixed resistance in order to obtain a higher efficiency of energy transfer.